

Hate Speech Classification and Reclaimed Language: Mitigating False Positives and Compounded Discrimination

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Content Warning

The following presentation contains use of and reference to racial, homophobic and transphobic slurs which viewers may find triggering. References to sentences containing such slurs are purely for illustration purposes and in no way reflect the author's attitudes or opinions.

CONTENTS

1.	Theoretical Background
2.	Research Question
3.	Quantifying the Problem
4.	Target Model
5.	Fine-tuning Attempt 1
6.	Balanced Dataset Creation
7.	Fine-tuning Attempt 2
8.	Model Performance
9.	Key Findings
10.	Limitations
11	Future Work

THEORETICAL BACKGROUND

- Hate-speech classification models have not been adapted to handle instances of “reclaimed language” (Thiago et al., 2021).
- Example: “Sissy”

	Hateful	Reclaimed
Definition	<i>Disparaging and Offensive. An effeminate boy or man.</i>	<i>To walk with sass. Contestants on RuPaul’s Drag Race are often given this advice by RuPaul before their runway walk .</i>
Usage	<i>“Look, that boy’s crying! What a sissy!”</i>	<i>“Sissy That Walk, queens!”</i>

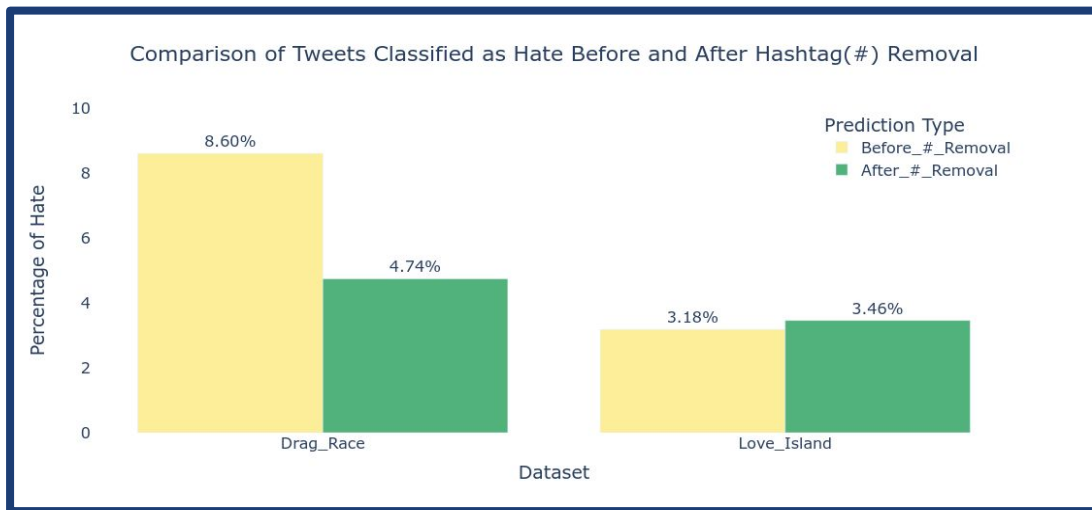
RESEARCH QUESTION

Can a hate speech classification model be fine tuned to reduce false positive classifications of reclaimed language whilst maintaining its performance in standard hate-speech classification tasks?

QUANTIFYING THE PROBLEM

Token	Percentage of Dataset Where Token Was Highest Contributing to Classification
Drag	30.5
Race	6.05
</s>	5.12
Gbitch	4.19
I	2.56
Gfucking	2.33
race	2.09
GI	2.09
Gass	1.63
.	1.40

Table showing Top 10 highest contributing tokens in hate-speech classification of tweets containing #DragRace



Before Removing Terms
"#DragRace" and
"#LoveIsland"



After Removing Terms
"#DragRace" and
"#LoveIsland"

TARGET MODEL: VIDGEN ET AL. (2020)

- RoBERTa model fine-tuned on complex perturbations of hate-speech and adversarial examples
- Four training rounds with human-in-the-loop synthetic hate speech generation
- **72.93±0.56 Macro F1** on most complex perturbation round
- 95% accuracy for hateful entries, and **93% non-hateful entries on “HATECHECK”** test suite



Performance of target models over four rounds of training on HATECHECK test suite

Pivot Type	Example Sentence
(HATE) Terms with positive sentiment	<i>'I just love it when all of the brown people leave my office so I don't have to look at them'.</i>
(NOT HATE) Not hateful use of a term that is also a referent for an identity (polysemy)	<i>'the black cat down the road really ruins the neighbourhood'.</i>

Example “pivots” used to create complex perturbations of hate-speech by Vidgen et al. (2021)

FINE-TUNING ATTEMPT 1: UNBALANCED DATASET

Training Dataset

- ~1000 sentences of transcript from the RuPaul's Drag TV show, which the model had previously misclassified as hateful.

Results

- **Perfect** accuracy for *Drag Race* transcript sentences
- **F1 reduced from** ~73% to 40.1% on Vidgen et al (2020) Training Round 4 Test Split

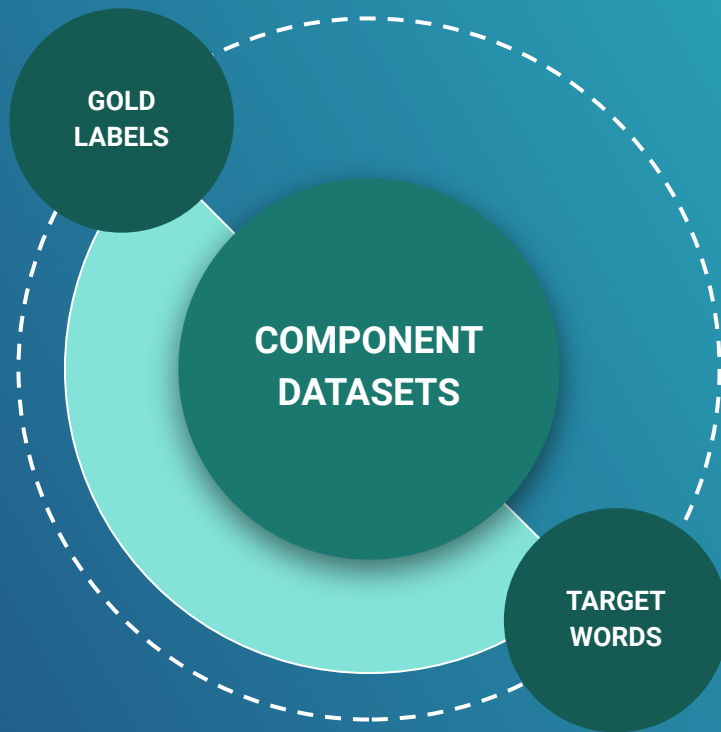
Conclusion

- “Catastrophic forgetting”
- Need for balanced dataset

CHARACTERISTICS AND HYPERPARAMETERS

Base Model	Vidgen et al.'s (2020) fine-tuned RoBERTa
Training Loop	Hugging Face Trainer Class
Batch size	16
Epochs	3
Optimizer	Adam
Learning Rate	0.0005.

BALANCED DATASET CREATION



GOLD LABELS

Gold Label	Proportion of Dataset (%)
DEG	58.4
NDG	33.5
APR	7.04
HOM	0.47
APR_negimp	0.25
CMP	0.24
APR_negexp	0.10

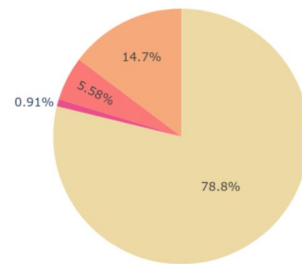
LABEL	INFO
DEG	Derogatory
NDG	Non Derogatory Non Appropriative
HOM	Homonym
APR	Appropriative
CMP	Noise

TARGET WORDS

Target Word	Proportion Hateful (%)	Proportion Reclaimed (%)	Proportion of Dataset Containing Word (%)
fag	59.3	2.20	67.8
bitch	65.2	14.7	12.6
gay	28.8	5.77	8.37
puss	72.3	3.72	2.92
homo	20.0	1.88	2.89
other	0.26	99.7	1.88
queer	29.8	8.26	1.08
drag	13.3	74.7	0.82
whore	63.0	18.1	0.68
queen	33.3	55.2	0.47
slut	55.4	8.93	0.28
sissy	44.8	3.45	0.14
slay	16.7	66.7	0.06

COMPONENT DATASETS

Distribution of Component Datasets in Final Dataset



FINE-TUNING ATTEMPT 2: BALANCED DATASET

Dataset and Training

- Train/Validation/Test Split from balanced dataset
- 70:15:15
- Stratified by Gold Label
- Attempted Cross-Validation ($k = 5$) to produce fine-tuned model by:
 - Best Fold
 - Averaged Weights

Results

- Improvement in **both accuracy and F1** scores
- Best model selected: without cross-validation
- Named **“Reclaimed Hate Speech” (RHS) Model**
 - *Due to performance on Test Suite*

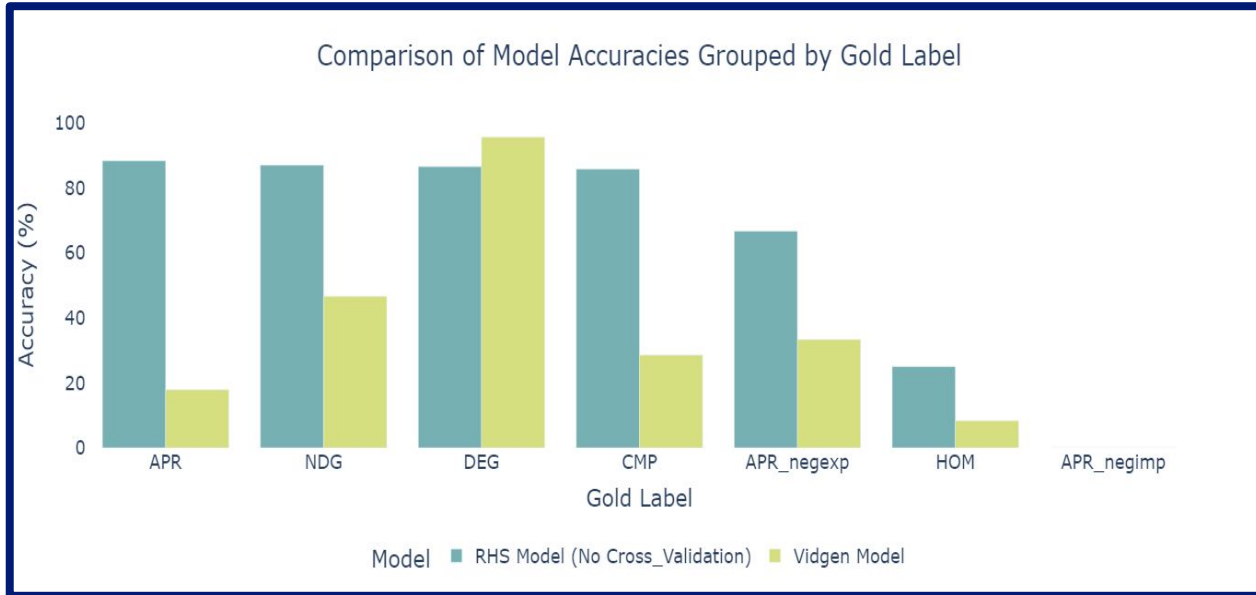
Reclaimed Hate Speech (RHS) Model Performance

	Accuracy (%)	F1 (%)
Test Dataset		
RHS	86.3	88.0
Vidgen et al. (2020)	72.7	80.4

CHARACTERISTICS AND HYPERPARAMETERS

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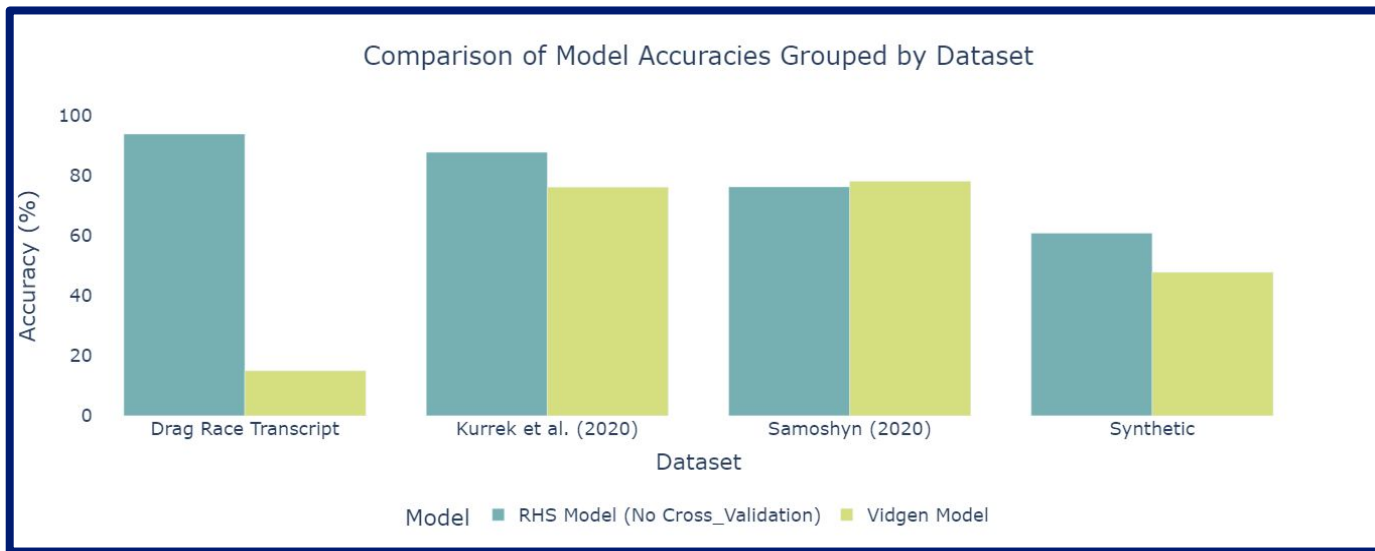
RHS MODEL PERFORMANCE: GOLD LABEL LEVEL



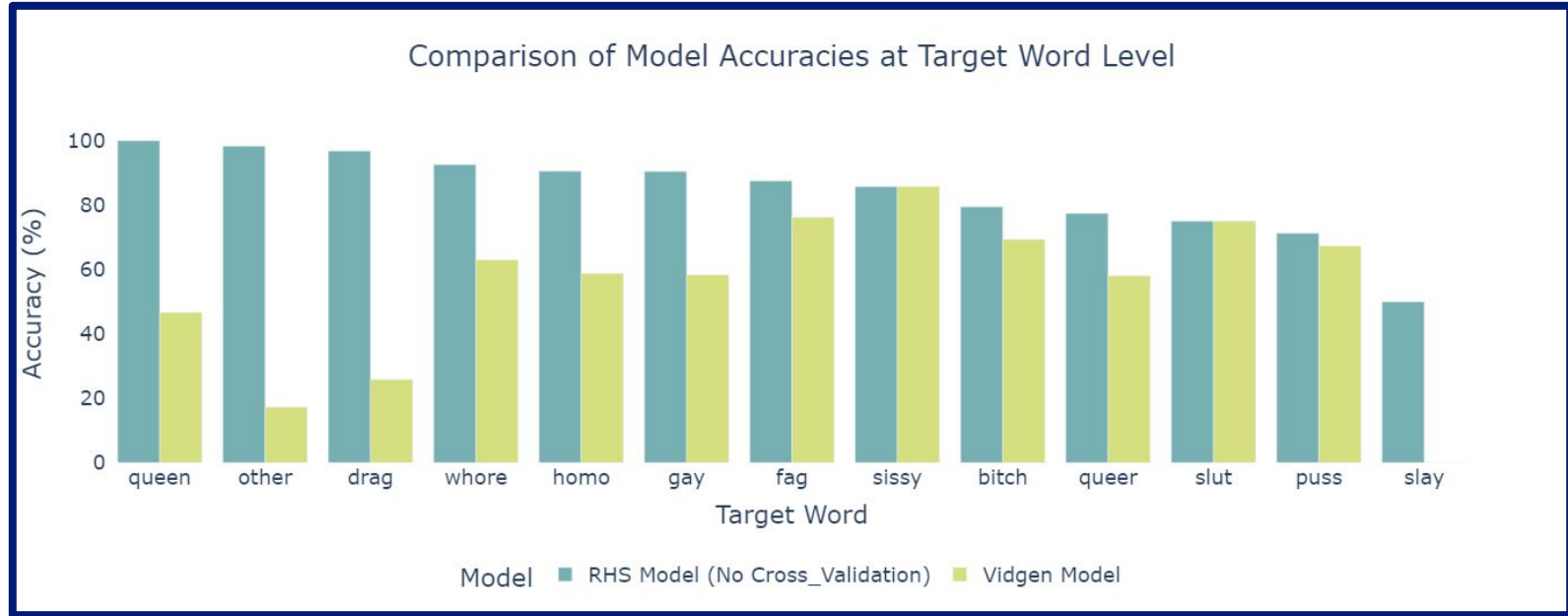
LABEL	INFO
DEG	Derogatory
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**NB: “APR” =
RECLAIMED**

RHS MODEL PERFORMANCE: DATASET LEVEL



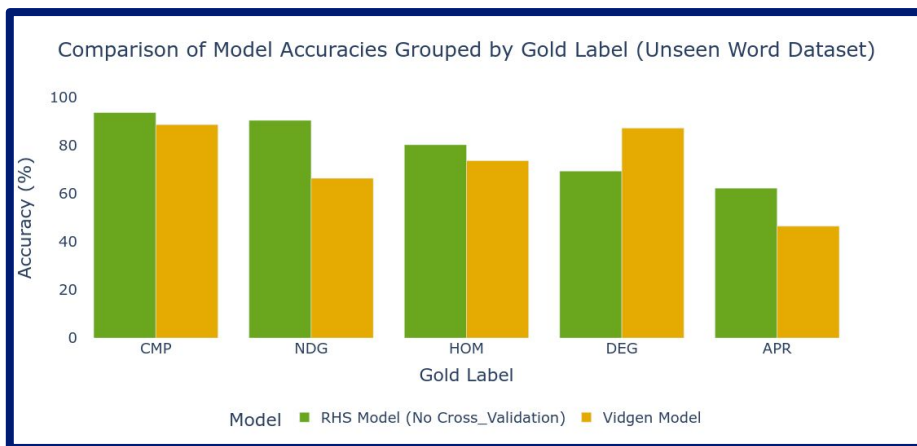
RHS MODEL PERFORMANCE: TARGET WORD LEVEL



RHS MODEL PERFORMANCE: HATECHECK TEST SUITE

HateCheck Validation Dataset		
	Accuracy (%)	F1 (%)
RHS (No Cross-Validation)	91.5	91.6
RHS (Best Fold)	73.9	74.8
RHS (Averaged Weights)	84.8	85.3
Vidgen et al. (2020)	95.6	95.6

RHS MODEL PERFORMANCE: UNSEEN TARGET WORD



	Accuracy (%)	F1 (%)
Unseen Target Word Dataset		
RHS	76.3	77.0
Vidgen et al. (2020)	79.1	82.7



KEY FINDINGS

13.7%

Increase in accuracy

7.6%

Increase in F1

On Test Dataset



- ◆ Compared to Vidgen et al.'s (2020) model performance

78.9%

Increase in accuracy in
classifying drag queen
transcript as not hateful



- ◆ Increased accuracy in 3 of 4 datasets
- ◆ Comparable performance in Hate Speech and Toxicity Dataset

>90%

Accuracy and F1 scores
in HATECHECK Test Suite



- Compared to Vidgen et al. (2020):
- 4.1% decrease in accuracy
 - 4.0% decrease in F1

LIMITATIONS

- *Generalisability*
- *Binary Hate Speech Classification*
- *Decreased Performance in Classifying Derogatory Sentences*
- *Synthetic Data Performance*
- *Unseen Target Word Performance*
- *Cross-Validation Model Performances*

FUTURE WORK

- *Engaging LGBTQ+ Community*
- *Optimising Balanced Dataset*
- *Third Classification Category (Offensive)*
- *Expanding Synthetic Dataset*
- *Targeted Improvement of Hate Detection*

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